DNS, WHOIS, and Email Header Analysis

Internship Assignment

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Date: March 5, 2026

Contents

1. Objective

The goal of this assignment is to get hands-on experience with three essential tools used in cybersecurity investigations—DNS lookups, WHOIS queries, and email header analysis. Taken together, these techniques allow an analyst to trace where an email came from, verify whether its claimed identity is genuine, and cross-check the infrastructure behind a domain.

This kind of analysis is a core part of phishing detection and email-based threat investigations. By walking through a real example with a test email, the aim is to build intuition about how each tool contributes to the bigger picture.

2. Setting Up the Test Email

To have something concrete to analyse, a test email was created and sent between two accounts under my control.

Field	Value
Sender	`hacker_scammer1@proton.me`
Receiver	`prajwalsb041@gmail.com`
Subject	Email Header Analysis Test

A ProtonMail account was chosen as the sender because it uses privacy-preserving infrastructure, which makes it an interesting case for header analysis—real attackers often choose services like ProtonMail for the same reason. The email was received in Gmail, which makes the full header easy to extract.

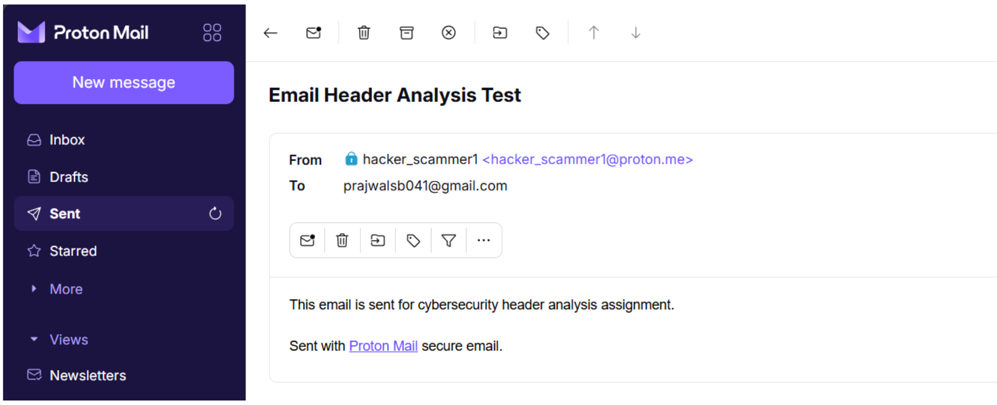


Figure 1: Test email generated from proton mail.

3. Extracting the Email Header from Gmail

Gmail hides the full technical header by default, but it can be revealed with a few clicks. Here is how it was done:

Step 1. Open the received email in Gmail.

Step 2. Click the three-dot menu (⋮) next to the Reply button (top-right of the message).

Step 3. Select **Show Original** from the dropdown.

Step 4. A new tab opens containing the complete raw email header.

The header is essentially a timestamped log of every mail server the message passed through, along with authentication results and metadata the sending server attached before dispatch.

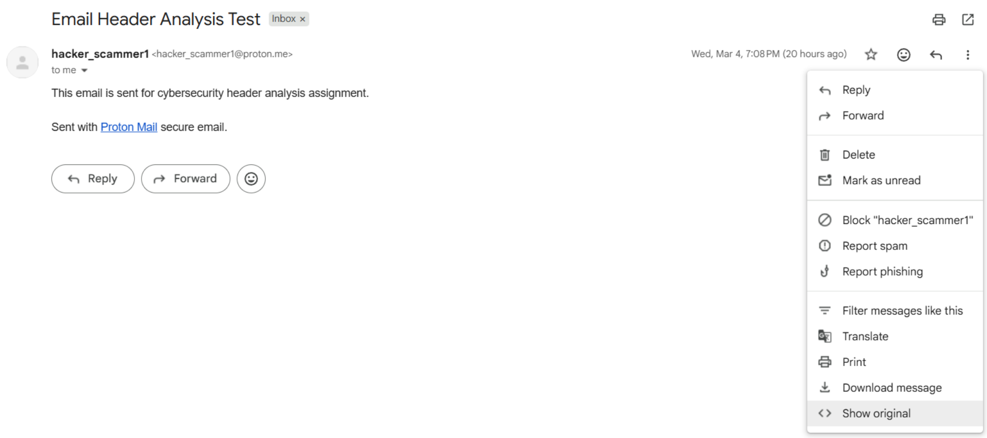


Figure 2: Gmail Show Original option.

```
ARC-Seal: i=1; a=rsa-sha256; t=1772631534; cv=none;
        d=google.com; s=arc-20240605;
        b=gcinFftxMG9frhMnbCWHADUXyCN9t78ZYdRlYXC5PEitf6vDAcVacfYbs30cH8FUeL
         sOqmklkhic7O3+QwPHuKNO/0hBOu2KcUmUSTrGksX+6GGf9ufiErg+uScKuHYORBMUFK
         CQY3colPwkjFKdbBlyjXpp4FZGwd1aV0xSJkIoUlMUHPydIWshxOdomkPkzJMokYSDi2
         mlZFDU5CvcT5OMLgqFw1o/gvLhvHfg8vqAOimfcmgi42wYCi4kI4sakvSOxQTuCDHCd/
         kUvGk6G9Ql9riA/KbN6nr4AZMgIIk2JUFhMvvbMow+w5vGeRAYGexxmxqjUkPtci8bUf
         WzGw==
ARC-Message-Signature: i=1; a=rsa-sha256; c=relaxed/relaxed; d=google.com; s=arc-20240605;
        h=mime-version:feedback-id:message-id:subject:from:to:date
         :dkim-signature;
        bh=JLFPQODz5FGrkQwgNwEsn0yaNLCWk2gcP73c6XLSdwI=;
        fh=pgT6vOR9V2FYJrt4L2vRIVBF7N7NXt0Jnuzoz5dfuQ8=;
        b=Y24NOY4KZegodTHWfdzVkWn1/NwxgEAoL8bCcOXM/XuaeqSsdPhSAxlNS2c3TlAFjj
         Gi97ifcmp1mlNZgNpsiPhZPSYKp5oT7u7JAiewJ9jgZF8+czSQpxo47inx272T59CQhu
         Udy/IIAaOXj2fWjxdgtvmiBfQY/QpRw88vCeM6Z2nmd7CrQzqsFGYFwBGRFNK9BO4VvN
         TGnbjfcz+1rcqcQS/hZSWqMRePCF9kcIO2uplWfFYHtOFzR1zz6j1x+mWdxUtAdobeSy
         N/isJL7zccI7szsueJJhngZFZU97KY8dmIo9lwtttXDFEZErRgt0S/OVsy8XKrYjZs+5
         ZsnQ==;
        dara=google.com
ARC-Authentication-Results: i=1; mx.google.com;
       dkim=pass header.i=@proton.me header.s=protonmail header.b=SlVH41RC;
       spf=pass (google.com: domain of hacker_scammer1@proton.me designates 185.70.43.167 as permitted sender)
smtp.mailfrom=hacker_scammer1@proton.me;
       dmarc=pass (p=QUARANTINE sp=QUARANTINE dis=NONE) header.from=proton.me
Return-Path: <hacker_scammer1@proton.me>
Received: from mail-43167.protonmail.ch (mail-43167.protonmail.ch. [185.70.43.167])
        by mx.google.com with ESMTPS id ffacd0b85a97d-439c65e1fcesi5041699f8f.28.2026.03.04.05.38.54
        for <prajwalsb041@gmail.com>
        (version=TLS1_3 cipher=TLS_AES_256_GCM_SHA384 bits=256/256);
        Wed, 04 Mar 2026 05:38:54 -0800 (PST)
Received-SPF: pass (google.com: domain of hacker_scammer1@proton.me designates 185.70.43.167 as permitted sender) client-ip=185.70.43.167;
Authentication-Results: mx.google.com;
       dkim=pass header.i=@proton.me header.s=protonmail header.b=SlVH41RC;
       spf=pass (google.com: domain of hacker_scammer1@proton.me designates 185.70.43.167 as permitted sender)
smtp.mailfrom=hacker_scammer1@proton.me;
       dmarc=pass (p=QUARANTINE sp=QUARANTINE dis=NONE) header.from=proton.me
DKIM-Signature: v=1; a=rsa-sha256; c=relaxed/relaxed; d=proton.me; s=protonmail; t=1772631534; x=1772890734;
bh=JLFPQODz5FGrkQwgNwEsn0yaNLCWk2gcP73c6XLSdwI=; h=Date:To:From:Subject:Message-ID:Feedback-ID:From:To:Cc:Date:
         Subject:Reply-To:Feedback-ID:Message-ID:BIMI-Selector; b=SlVH41RC4aT9CiMTEg2zrvA1OIGEMeVEYfIq/ldDHqz5yRqHH3f6fERfkNJbpivoi
         rhFeRnldJp/mdJ8A2jbpgPo/GptWTw9asLMSMhUe9A+Dnq4/IzLX/R989aNLwB5c5n
         4b9U/NtkX51VD+/1DGhUbLR21KXaMrFvNAaWIDlTlTiLq3a9hoTcfLaueWMVxjRbEm
         cRXiuAZxUrFtwlmaLFE3bJ6rkmsHFhnhG7LyvkFwYJcpLrPiEvN2nrm9cRNnHeccwc
         Ioq2H7elRwXU2lOX3Ty8C1tlBnHpKAZZBBUShKuAAwmo2JTsRxI1ivp8BXgH52y2q1
         LB+N0diG3YSfQ==
```

Figure 3: Raw email header.

3.1 Key Findings from the Header

After reading through the header, the following pieces of information stood out:

Field	Value
Sender mail server	`mail-43167.protonmail.ch`
Sender IP address	`185.70.43.167`
SPF result	PASS
DKIM result	PASS
DMARC result	PASS

All three authentication checks passing is a strong indicator that the email genuinely came from ProtonMail's servers—it had not been spoofed or tampered with in transit.

4. Email Header Analysis with MXToolbox

While reading the raw header manually is useful, specialised tools make the process much faster. MXToolbox's Email Header Analyzer was used to visualise the routing path and highlight any potential issues.

Tool used: `https://mxtoolbox.com/EmailHeaders.aspx`

4.1 Steps Performed

Step 1. Copied the full raw header from Gmail's Show Original view.

Step 2. Navigated to the MXToolbox Email Header Analyzer page.

Step 3. Pasted the header into the input box and clicked **Analyze Header**.

Step 4. Reviewed the results, including the relay hops, timestamps, and authentication summary.

MXToolbox lays out each hop in the email's journey as a clear table, showing how long the message spent at each server. This is particularly useful when investigating delayed or suspicious emails.

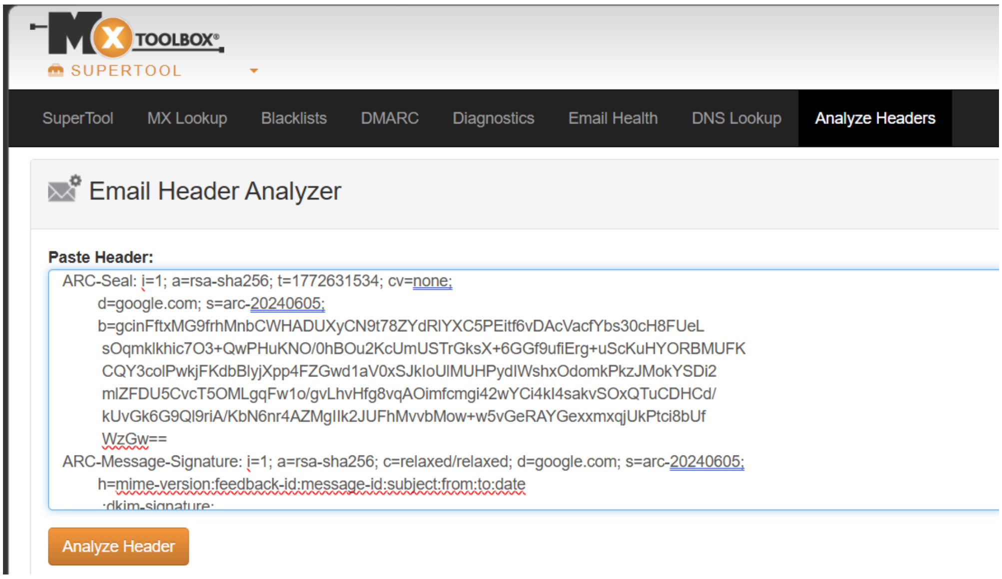


Figure 4: MXToolbox - pasting the raw header.

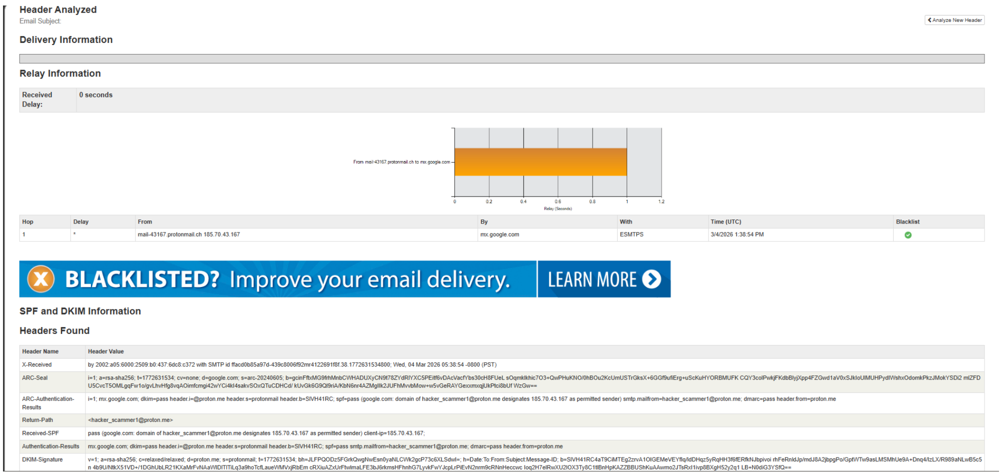


Figure 5: MXToolbox analysis results.

5. DNS Investigation

DNS (Domain Name System) records hold publicly accessible information about a domain—who handles its email, where its servers are, and who manages its name resolution. Several queries were run against `proton.me` to build a picture of its infrastructure.

5.1 Basic Domain Lookup (nslookup)

```
nslookup proton.me
```

This is the most straightforward DNS query—it asks: *"What IP address does this domain name point to?"* The response confirms that `proton.me` resolves to a real address and gives a first indication of where the web server lives.

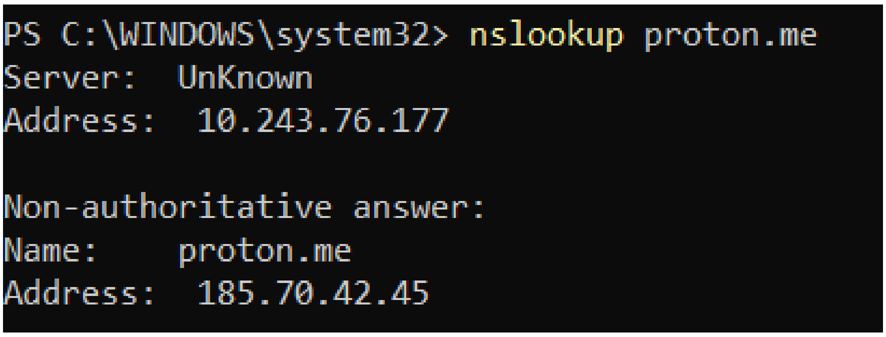


Figure 6: Output of nslookup proton.me.

5.2 IPv4 Address Record (A Record)

```
Resolve-DnsName proton.me -Type A
```

An **A record** maps a hostname to a 32-bit IPv4 address. This query was used to confirm the exact IPv4 address assigned to `proton.me`, which can later be cross-referenced with WHOIS data to verify ownership.

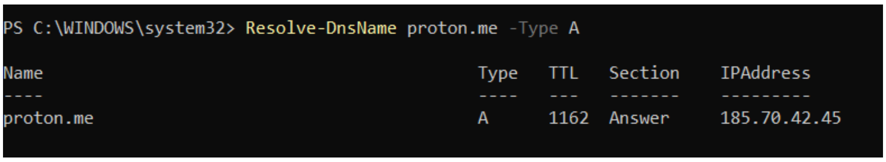
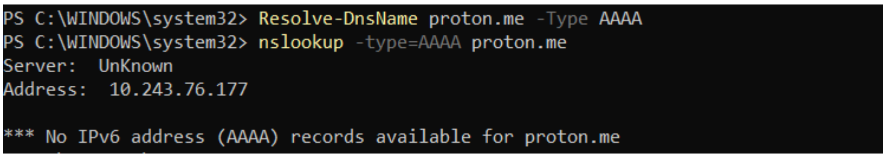


Figure 7: A record query result.

5.3 IPv6 Address Record (AAAA Record)

```
Resolve-DnsName proton.me -Type AAAA
```

An **AAAA record** does the same as an A record but for 128-bit IPv6 addresses. This query returned no results, which simply means that `proton.me` does not advertise an IPv6 address through the DNS server that was queried. This is not unusual—many services still operate solely on IPv4.



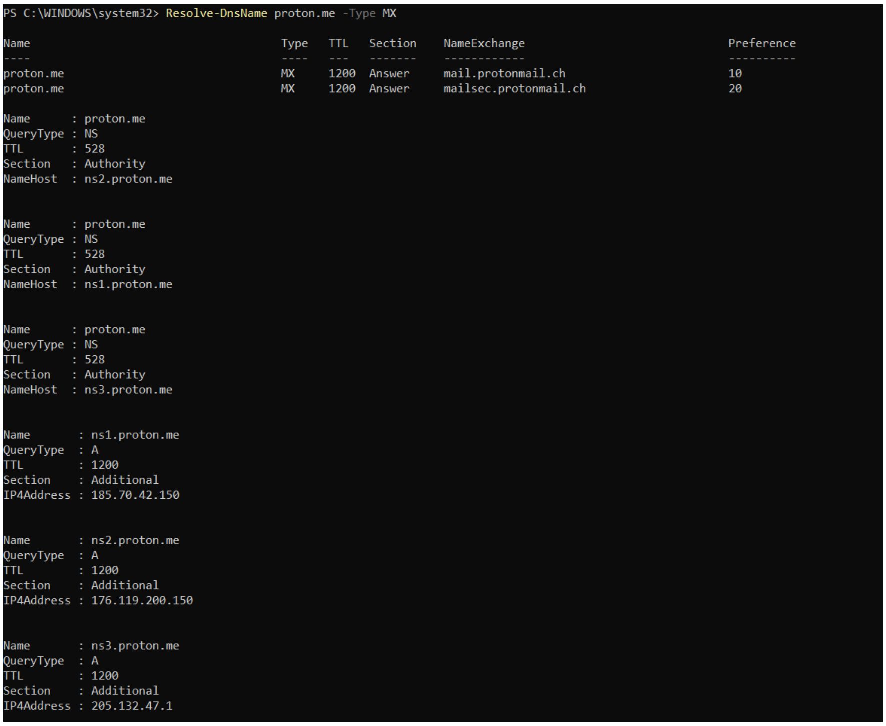
```
PS C:\WINDOWS\system32> Resolve-DnsName proton.me -Type AAAA
PS C:\WINDOWS\system32> nslookup -type=AAAA proton.me
Server:  UnKnown
Address:  10.243.76.177

*** No IPv6 address (AAAA) records available for proton.me
```

Figure 8: AAAA record query result.

5.4 Mail Server Records (MX Records)

```
Resolve-DnsName proton.me -Type MX
```

MX (Mail Exchanger) records specify which servers are authorised to accept incoming email for a domain. Checking these records helps confirm whether the mail server identified in the email header is actually listed as a legitimate mail handler for that domain—a quick way to spot spoofing attempts.



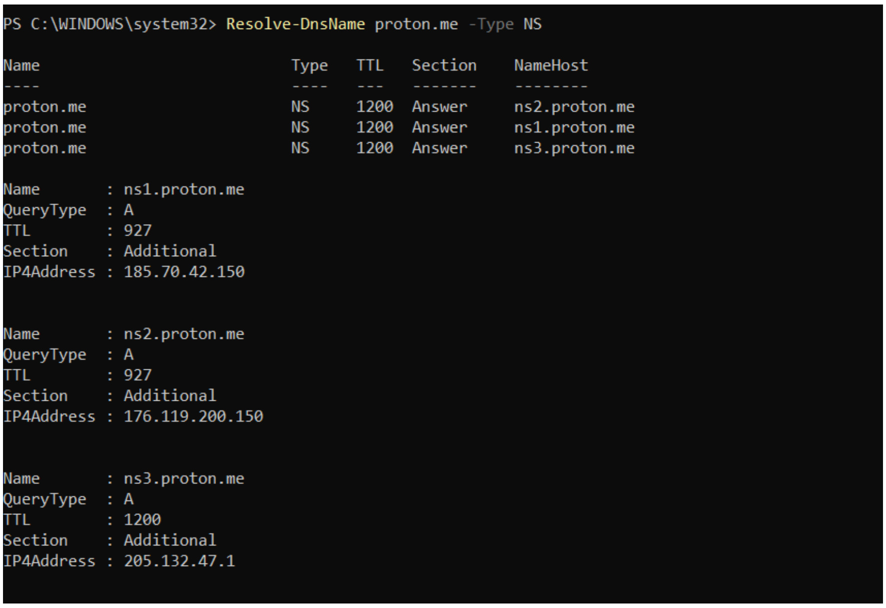
```
PS C:\WINDOWS\system32> Resolve-DnsName proton.me -Type MX

Name                                     Type   TTL   Section    NameExchange                              Preference
----                                     ----   ---   -------    ------------                              ----------
proton.me                                MX     1200  Answer     mail.protonmail.ch                        10
proton.me                                MX     1200  Answer     mailsec.protonmail.ch                     20

Name      : proton.me
QueryType : NS
TTL       : 528
Section   : Authority
NameHost  : ns2.proton.me


Name      : proton.me
QueryType : NS
TTL       : 528
Section   : Authority
NameHost  : ns1.proton.me


Name      : proton.me
QueryType : NS
TTL       : 528
Section   : Authority
NameHost  : ns3.proton.me


Name       : ns1.proton.me
QueryType  : A
TTL        : 1200
Section    : Additional
IP4Address : 185.70.42.150


Name       : ns2.proton.me
QueryType  : A
TTL        : 1200
Section    : Additional
IP4Address : 176.119.200.150


Name       : ns3.proton.me
QueryType  : A
TTL        : 1200
Section    : Additional
IP4Address : 205.132.47.1
```

Figure 9: MX record query result.

5.5 Name Server Records (NS Records)

```
Resolve-DnsName proton.me -Type NS
```

NS (Name Server) records point to the authoritative DNS servers for a domain—the servers that hold the definitive records and answer queries about that domain. Knowing who manages a domain's DNS is useful when tracing the origin of a suspicious domain.



```
PS C:\WINDOWS\system32> Resolve-DnsName proton.me -Type NS

Name                           Type   TTL   Section    NameHost
----                           ----   ---   -------    --------
proton.me                      NS     1200  Answer     ns2.proton.me
proton.me                      NS     1200  Answer     ns1.proton.me
proton.me                      NS     1200  Answer     ns3.proton.me

Name       : ns1.proton.me
QueryType  : A
TTL        : 927
Section    : Additional
IP4Address : 185.70.42.150


Name       : ns2.proton.me
QueryType  : A
TTL        : 927
Section    : Additional
IP4Address : 176.119.200.150


Name       : ns3.proton.me
QueryType  : A
TTL        : 1200
Section    : Additional
IP4Address : 205.132.47.1
```

Figure 10: NS record query result.

6. WHOIS Lookup

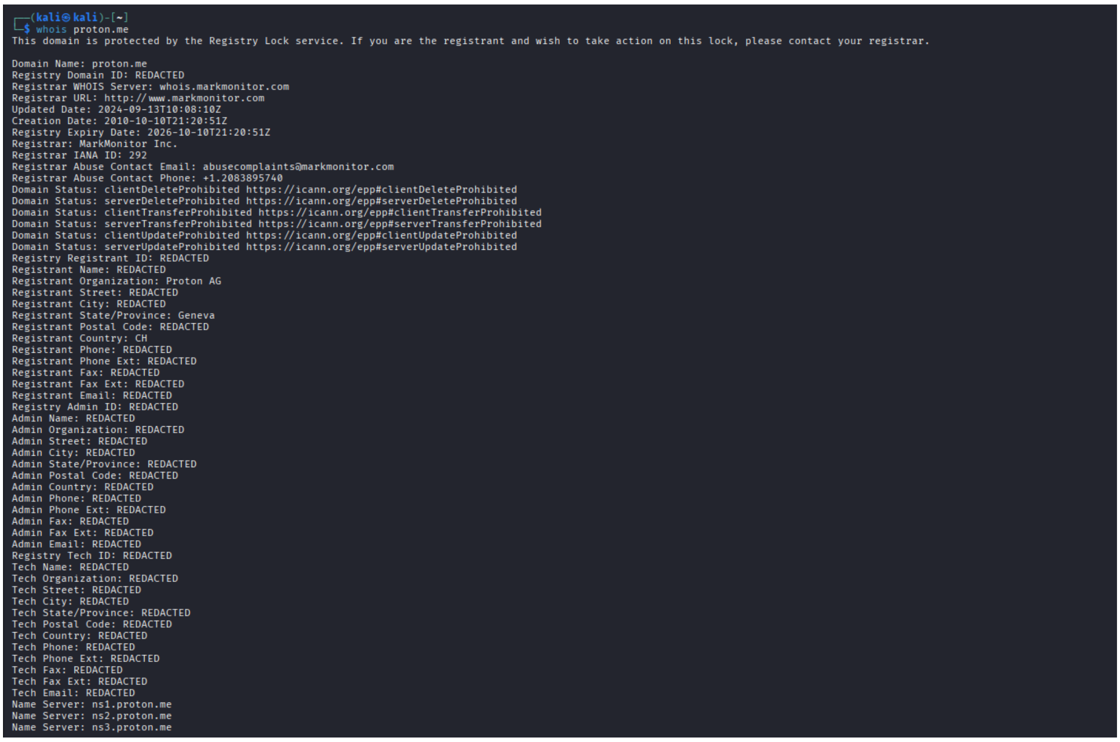
WHOIS is a protocol that retrieves registration information about a domain directly from the registrar's database. It is one of the first tools a security analyst reaches for when trying to determine whether a domain is legitimate.

```
whois proton.me
```

The WHOIS record for a domain typically contains:

- **Registrar** — the company where the domain was purchased.
- **Creation date** — when the domain was first registered (a brand-new domain associated with financial requests is a red flag).
- **Expiry date** — when the registration runs out.
- **Name servers** — consistent with the DNS NS records found earlier.
- **Registrant organisation** — who owns the domain (may be redacted for privacy).

For `proton.me`, the WHOIS data confirmed the domain is registered and maintained by Proton AG, the Swiss company behind ProtonMail. This is exactly what would be expected for a legitimate email service.



```
┌──(kali㉿kali)-[~]
└─$ whois proton.me
This domain is protected by the Registry Lock service. If you are the registrant and wish to take action on this lock, please contact your registrar.

Domain Name: proton.me
Registry Domain ID: REDACTED
Registrar WHOIS Server: whois.markmonitor.com
Registrar URL: http://www.markmonitor.com
Updated Date: 2024-09-13T10:08:10Z
Creation Date: 2010-10-10T21:20:51Z
Registry Expiry Date: 2026-10-10T21:20:51Z
Registrar: MarkMonitor Inc.
Registrar IANA ID: 292
Registrar Abuse Contact Email: abusecomplaints@markmonitor.com
Registrar Abuse Contact Phone: +1.2083895740
Domain Status: clientDeleteProhibited https://icann.org/epp#clientDeleteProhibited
Domain Status: serverDeleteProhibited https://icann.org/epp#serverDeleteProhibited
Domain Status: clientTransferProhibited https://icann.org/epp#clientTransferProhibited
Domain Status: serverTransferProhibited https://icann.org/epp#serverTransferProhibited
Domain Status: clientUpdateProhibited https://icann.org/epp#clientUpdateProhibited
Domain Status: serverUpdateProhibited https://icann.org/epp#serverUpdateProhibited
Registry Registrant ID: REDACTED
Registrant Name: REDACTED
Registrant Organization: Proton AG
Registrant Street: REDACTED
Registrant City: REDACTED
Registrant State/Province: Geneva
Registrant Postal Code: REDACTED
Registrant Country: CH
Registrant Phone: REDACTED
Registrant Phone Ext: REDACTED
Registrant Fax: REDACTED
Registrant Fax Ext: REDACTED
Registrant Email: REDACTED
Registry Admin ID: REDACTED
Admin Name: REDACTED
Admin Organization: REDACTED
Admin Street: REDACTED
Admin City: REDACTED
Admin State/Province: REDACTED
Admin Postal Code: REDACTED
Admin Country: REDACTED
Admin Phone: REDACTED
Admin Phone Ext: REDACTED
Admin Fax: REDACTED
Admin Fax Ext: REDACTED
Admin Email: REDACTED
Registry Tech ID: REDACTED
Tech Name: REDACTED
Tech Organization: REDACTED
Tech Street: REDACTED
Tech City: REDACTED
Tech State/Province: REDACTED
Tech Postal Code: REDACTED
Tech Country: REDACTED
Tech Phone: REDACTED
Tech Phone Ext: REDACTED
Tech Fax: REDACTED
Tech Fax Ext: REDACTED
Tech Email: REDACTED
Name Server: ns1.proton.me
Name Server: ns2.proton.me
Name Server: ns3.proton.me
```

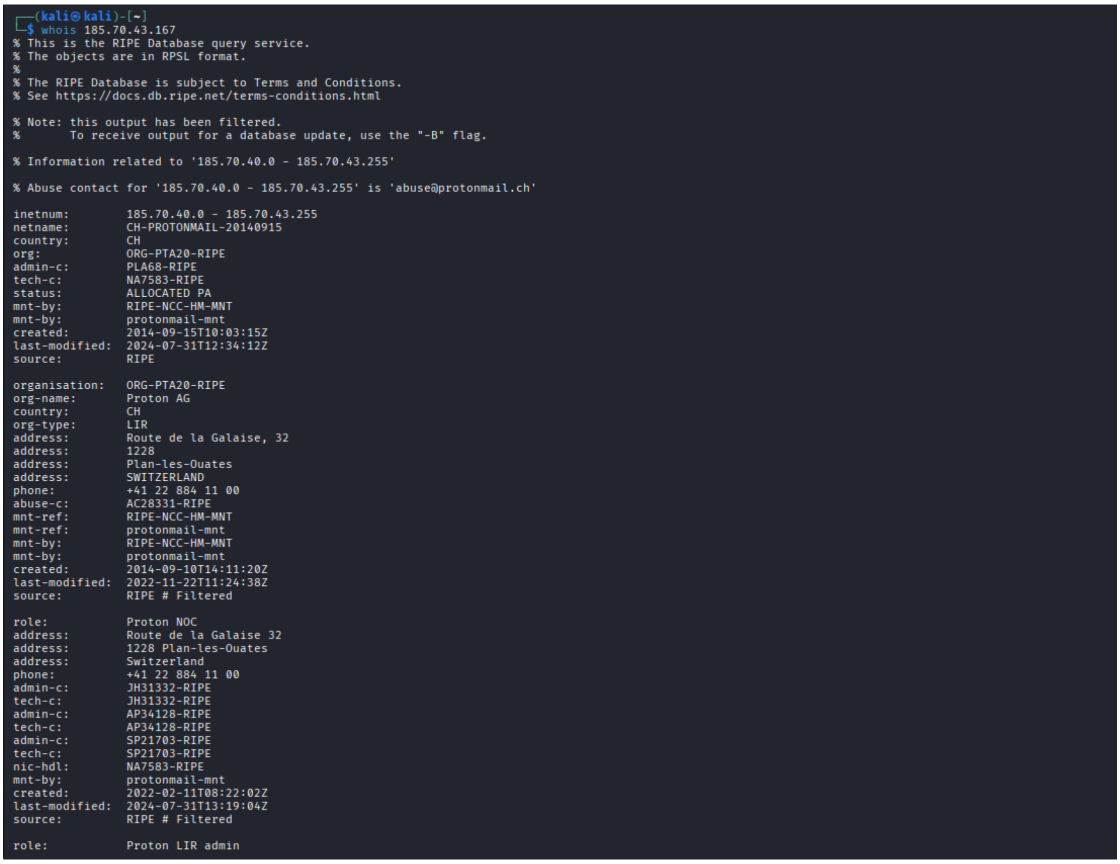
Figure 11: WHOIS lookup for proton.me.

7. IP Address Investigation

The email header identified the sending IP address as `185.70.43.167`. Performing a WHOIS lookup on an IP address (rather than a domain name) reveals which organisation has been allocated that block of addresses by the regional internet registry.

```
whois 185.70.43.167
```

The result confirmed that `185.70.43.167` belongs to ProtonMail's network allocation. This is consistent with the rest of the analysis—the email was sent from a genuine ProtonMail server, not from a third-party host trying to impersonate ProtonMail.



```
┌──(kali㉿kali)-[~]
└─$ whois 185.70.43.167
% This is the RIPE Database query service.
% The objects are in RPSL format.
%
% The RIPE Database is subject to Terms and Conditions.
% See https://docs.db.ripe.net/terms-conditions.html

% Note: this output has been filtered.
%       To receive output for a database update, use the "-B" flag.

% Information related to '185.70.40.0 - 185.70.43.255'

% Abuse contact for '185.70.40.0 - 185.70.43.255' is 'abuse@protonmail.ch'

inetnum:        185.70.40.0 - 185.70.43.255
netname:        CH-PROTONMAIL-20140915
country:        CH
org:            ORG-PTA20-RIPE
admin-c:        PLA68-RIPE
tech-c:         NA7583-RIPE
status:         ALLOCATED PA
mnt-by:         RIPE-NCC-HM-MNT
mnt-by:         protonmail-mnt
created:        2014-09-15T10:03:15Z
last-modified:  2024-07-31T12:34:12Z
source:         RIPE

organisation:   ORG-PTA20-RIPE
org-name:       Proton AG
country:        CH
org-type:       LIR
address:        Route de la Galaise, 32
address:        1228
address:        Plan-les-Ouates
address:        SWITZERLAND
phone:          +41 22 884 11 00
abuse-c:        AC28331-RIPE
mnt-ref:        RIPE-NCC-HM-MNT
mnt-ref:        protonmail-mnt
mnt-by:         RIPE-NCC-HM-MNT
mnt-by:         protonmail-mnt
created:        2014-09-10T14:11:20Z
last-modified:  2022-11-22T11:24:38Z
source:         RIPE # Filtered

role:           Proton NOC
address:        Route de la Galaise 32
address:        1228 Plan-les-Ouates
address:        Switzerland
phone:          +41 22 884 11 00
admin-c:        JH31332-RIPE
tech-c:         JH31332-RIPE
admin-c:        AP34128-RIPE
tech-c:         AP34128-RIPE
admin-c:        SP21703-RIPE
tech-c:         SP21703-RIPE
nic-hdl:        NA7583-RIPE
mnt-by:         protonmail-mnt
created:        2022-02-11T08:22:02Z
last-modified:  2024-07-31T13:19:04Z
source:         RIPE # Filtered

role:           Proton LIR admin
```

Figure 12: WHOIS lookup for IP 185.70.43.167.

8. Conclusion

Going through this exercise showed me how much you can uncover about an email without any special tools—just DNS, WHOIS, and the header itself. In this case, everything checked out: the email came from a real ProtonMail server, the authentication passed, and the domain ownership was legitimate.

But the same three checks on a malicious email would tell a very different story. That is what makes this worth knowing.